

Discussion

- Between the baseline model and the finetuned model, the baseline model actually performed better than the finetuned in my case. The difference in performance is mainly caused by the poor dataset I collected and used for finetuning.
- For the dataset collection, it was actually split into three person: me, Ben, and Adi. Although we were each supposed to collect 10 labeled numbers from 0-9, the combined dataset ended up with only 74 features, and 19 of those were unlabeled, which means the dataset is really small and skewed.
- Moreover, because the finetuning dataset was collected by three different individuals, that data wasn't tailored to me. Hence during deployment and inference, it wasn't really doing that well for my specific writing habits.

Baseline model

The screenshot shows the Arduino IDE interface with the 'Serial Monitor' window open. The top toolbar indicates the board is 'Arduino Nano 33 BLE'. The file explorer shows several files: 'magic_wand_lab7.ino', 'imu_provider.h', 'magic_wand_model_data.cpp', 'magic_wand_model_data.h', 'rasterize_stroke.cpp', and 'rasterize_stroke.h'. The code editor displays a line of hexadecimal data: '1308 0x80, 0xca, 0x13, 0x3a, 0xbe, 0xa3, 0xff, 0x39, 0x8c, 0x15, 0x00, 0x3a,'. The Serial Monitor window shows a message input field with the text 'Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')' and a 'New Line' button. The output area displays a series of dots and hash symbols, indicating the model's output. The status bar at the bottom shows 'Ln 1317, Col 61' and 'Arduino Nano 33 BLE on /dev/cu.usbmodem141101'.

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✓ → ↻

Arduino Nano 33 BLE

📶 🔊 🔍

magic_wand_lab7.ino imu_provider.h magic_wand_model_data.cpp magic_wand_model_data.h rasterize_stroke.cpp rasterize_stroke.h

1308 0x80, 0xca, 0x13, 0x3a, 0xbe, 0xa3, 0xff, 0x39, 0x8c, 0x15, 0x00, 0x3a,

Output Serial Monitor

Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')

New Line 115200 baud

Found 0 (127)

Ln 1317, Col 61 Arduino Nano 33 BLE on /dev/cu.usbmodem141101 2

✓ → ↻

Arduino Nano 33 BLE

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magic_wand_lab7.ino imu_provider.h magic_wand_model_data.cpp magic_wand_model_data.h rasterize_stroke.cpp rasterize_stroke.h

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Output Serial Monitor

Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')

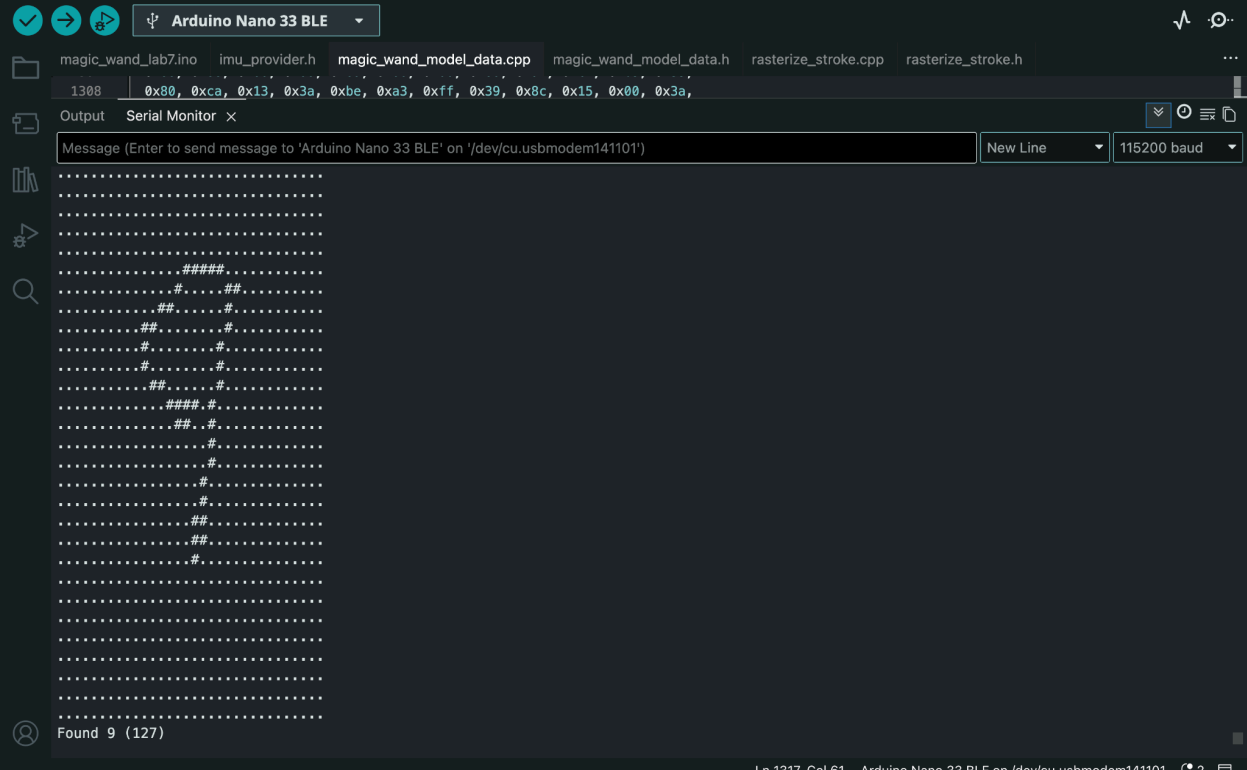
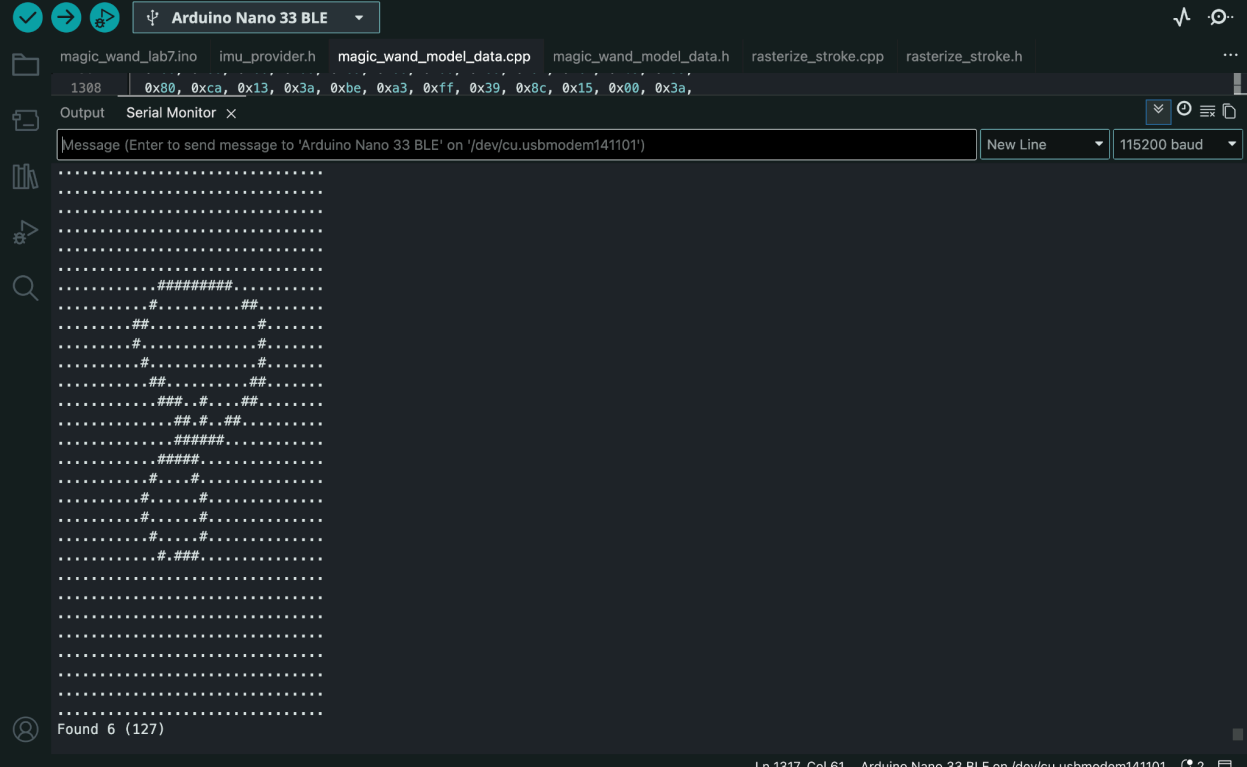
New Line 115200 baud

Found 3 (66)

Ln 1317, Col 61 Arduino Nano 33 BLE on /dev/cu.usbmodem141101 2

The screenshot shows the Arduino IDE interface. At the top, the board is set to "Arduino Nano 33 BLE". The file explorer on the left shows the project files: "magic_wand_lab7.ino", "imu_provider.h", "magic_wand_model_data.cpp", "magic_wand_model_data.h", "rasterize_stroke.cpp", and "rasterize_stroke.h". The main editor displays the "magic_wand_model_data.cpp" file, which contains a large array of hexadecimal values. The Serial Monitor window at the bottom is open, showing a message: "Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')". The monitor settings are set to "New Line" and "115200 baud". The status bar at the bottom indicates "Found 4 (127)" errors.

The screenshot shows the Arduino IDE interface with the Serial Monitor open. The top toolbar includes icons for checking, running, and uploading code. The file explorer on the left shows the project files: magic_wand_lab7.ino, imu_provider.h, magic_wand_model_data.cpp, magic_wand_model_data.h, rasterize_stroke.cpp, and rasterize_stroke.h. The Serial Monitor is set to 'New Line' and '115200 baud'. The output shows a series of hexadecimal values: 1308, 0x80, 0xca, 0x13, 0x3a, 0xbe, 0xa3, 0xff, 0x39, 0x8c, 0x15, 0x00, 0x3a, followed by a series of '#' characters. The status bar at the bottom indicates 'Found 5 (127)'.



Finetuned model

